Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-II-Q02] | Part A (Q1-Q5)

Answer: 098

1. The two factors $x-y, x+y$ have the same parity, and their product is even, so both are even. Set $x-y=2a, x+y=2b$ to get $ab=10^6=2^6 5^6$.

2. There are 49 ordered positive factor pairs. Each yields $x=a+b, y=b-a$; their simultaneous negatives give 49 more. Opposite signs cannot produce the positive product, so the total is 98.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-II-Q03] | Part A (Q1-Q5)

Answer: 111

1. The logarithms give $abc = 6^6$, and geometricity gives $abc = b^3$. Hence $b = 36$ and $ac = 1296$.

2. Write $a = 36 - k^2 > 0$ with $1 \le k \le 5$. The candidates are $35, 32, 27, 20, 11$; only 27 divides 1296.

3. Thus $(a, b, c) = (27, 36, 48)$, and the sum is 111.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q01] | Part A (Q1-Q5)

Answer: 336

1. Write the numbers as $a, b, c = a + b$. Then $abc = 12c$, so $ab = 12$ and $N = 12(a + b)$.

2. The unordered factor pairs $(1, 12), (2, 6), (3, 4)$ give $N = 156, 96, 84$. Their sum is 336.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q03] | Part A (Q1-Q5)

Answer: 384

1. For integer dimensions p, q, r, the hidden count is $(p-1)(q-1)(r-1) = 231 = 3 \cdot 7 \cdot 11$.

2. Up to order, the hidden dimensions are $(3, 7, 11), (1, 3, 77), (1, 7, 33), (1, 11, 21), (1, 1, 231)$. Adding one to each dimension gives total counts $384, 624, 544, 528, 928$.

3. The least is 384, from a $4 \times 8 \times 12$ block.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-I-Q04] | Part A (Q1-Q5)

Answer: 294

1. Write $N = s^2 + 5 = x(x+7)$. Completing the square gives $(2x+7-2s)(2x+7+2s) = 69$.

2. The positive factor pairs are $(1, 69)$ and $(3, 23)$. They give $(x, s) = (14, 17)$ or $(3, 5)$, hence sizes $294, 30$. The maximum is 294.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q03] | Part A (Q1-Q5)

Answer: 314

1. Eliminating b gives $4j + 5s = 57$. Its positive integer solutions are $(j, s) = (13, 1), (8, 5), (3, 9)$.

2. The respective $2b$ values are $31, 30, 29$, so integrality selects $(b, j, s) = (15, 8, 5)$. Squared sum $225 + 64 + 25 = 314$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q04] | Part A (Q1-Q5)

Answer: 080

1. Complete the square: $[y-(x+42)][y+(x+42)]=244$. Both factors are positive and of equal parity.

2. The only equal-parity positive factor pair is $2, 122$. Thus $y=62, x=18$, giving 80.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q04] | Part A (Q1-Q5)

Answer: 089

1. For c children, the arithmetic-series sum is $c(n+1-c)=2009$. Hence $c \mid 2009$ and $n=2009/c+c-1$. Last-place nonnegativity imposes $c(c-1) \leq 2009$.

2. Since $2009=7^2 \cdot 41$, allowed divisors are $c=1,7,41$; larger divisors already exceed the inequality. They give $n=2009,293,89$.

3. The minimum is 89, attained by forty-one children ending at nine grapes.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2010-I-Q05] | Part A (Q1-Q5)

Answer: 501

1. Factor the squared differences. Since $a-b, c-d \geq 1$, one has $(a-b)(a+b)+(c-d)(c+d) \geq a+b+c+d$, with equality only when both differences are 1.

2. Thus $b=a-1, d=c-1$ and $a+c=1006$. The strict middle inequality gives $a \geq 504$, while $d \geq 1$ gives $a \leq 1004$.

3. Every integer in that interval gives $(a, a-1, 1006-a, 1005-a)$, so the count is $1004-504+1=501$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2012-II-Q01] | Part A (Q1-Q5)

Answer: 034

1. Divide by four to get $5m + 3n = 503$. Modulo three, $m \equiv 1$, so write $m = 3k + 1, n = 166 - 5k$.

2. Positivity is exactly $0 \leq k \leq 33$, giving 34 pairs.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2021-I-Q05] | Part A (Q1-Q5)

Answer: 031

1. Write the terms as $a-d, a, a+d$ with $d \geq 1$. The equation is $d^2(a-2)=3a^2$, forcing $a > 2$.

2. With $t=a-2$, obtain $d^2=3t+12+12/t$, so t is a positive divisor of 12. For $t = 1,2,3,4,6,12$ the squared differences are $27,24,25,27,32,49$.

3. Only $(a,d)=(5,5),(14,7)$ work. Their last terms are $10, 21$, summing to 31.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2025-I-Q04] | Part A (Q1-Q5)

Answer: 117

1. Factor as $(3x+2y)(4x-3y)=0$. The two integer families are $(x,y)=(2t,-3t)$ and $(3t,4t)$.

2. The first permits $-33 \le t \le 33$, giving 67 pairs; the second permits $-25 \le t \le 25$, giving 51. Only the origin is shared, so total $67+51-1=117$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2026-I-Q04] | Part A (Q1-Q5)

Answer: 070

1. For $n = a + b + ab$, the condition is $n + 1 = (a + 1)(b + 1)$ with distinct factors at least two. Thus $n + 1$ must be composite but not a prime square.

2. Among $2, \ldots, 101$ there are 100 integers, 26 primes, and four prime squares $4, 9, 25, 49$. Every other composite has unequal factors obtained from its smallest prime factor. Count $100 - 26 - 4 = 70$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-I-Q06] | Part B (Q6-Q10)

Answer: 997

1. The equation gives $(\sqrt{y}-\sqrt{x})^2=4$, so $\sqrt{y}-\sqrt{x}=2$. Also $\sqrt{y}+\sqrt{x}=(y-x)/2$ is rational, making both square roots rational and hence integral.

2. Therefore $x=a^2, y=(a+2)^2$. The strict upper bound requires $a+2\leq 999$, so $1\leq a\leq 997$ and there are 997 pairs.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-I-Q08] | Part B (Q6-Q10)

Answer: 748

1. With first terms $x \leq y$, iteration gives $a_9 = 13x + 21y$. Suppose two distinct pairs have $x < u$; equality of ninth terms gives $13(u - x) = 21(y - v)$.

2. Coprimality implies $u - x \geq 21$, hence $u \geq 22$ and $v \geq u \geq 22$. Thus the common value is at least $34 \cdot 22 = 748$.

3. The pairs $(1, 35)$ and $(22, 22)$ both give 748, proving this lower bound attainable.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q07] | Part B (Q6-Q10)

Answer: 380

1. The altitude from D meets AC at its midpoint, distance 15 from A and 6 from B. If $a = AD, b = BD$, subtracting Pythagorean equations gives $(a+b)(a-b) = 225 - 36 = 189$.

2. The positive factor pairs yield $(a, b) = (95, 94), (33, 30), (17, 10), (15, 6)$. The last has zero altitude and is excluded.

3. The valid perimeters are $220, 96, 64$, summing to 380.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q08] | Part B (Q6-Q10)

Answer: 129

1. Write them as $a, a+d, a+2d, a+30$. Geometricity gives $(a+d)(a+30)=(a+2d)^2$, hence $3a(10-d)=2d(2d-15)$.

2. Positivity makes the two parenthesized factors have the same sign, so $7.5<d<10$. For $d=8$, $a=8/3$ is not integral; for $d=9$, $a=18$.

3. The sequence is $18, 27, 36, 48$, whose sum is 129.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q10] | Part B (Q6-Q10)

Answer: 156

1. Write them $x, x+60$. Squaring shows $x(x+60)=z^2$ for an integer z, hence $(x+30-z)(x+30+z)=900$.

2. The factors must both be even. Positive factor pairs below the equal pair give $x=196, 48, 20, 4$; the equal pair gives $x=0$, excluded.

3. At $x=196$, both original integers are squares and the resulting radicand is 900, forbidden. At $x=48$, their square roots sum to $10\sqrt{3}=\sqrt{300}$, allowed. Therefore the maximum sum is $48+108=156$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q06] | Part B (Q6-Q10)

Answer: 408

1. Integral splits force the three portions to be $8x, 8y, 24z$. The final totals are $6x + 3y + 11z$, $x + 2y + 11z$, $x + 3y + 2z$.

2. Equating these to ratio $3 : 2 : 1$ gives $3x + 5z = 6y$ and $x + 4y = 7z$. Elimination yields $9y = 13z$, hence $(x, y, z) = (11t, 13t, 9t)$ for positive integer t.

3. The initial total is $8x + 8y + 24z = 408t$, whose minimum is 408.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q09] | Part B (Q6-Q10)

Answer: 000

1. Subtract one from each variable in the first equation. It bijects to nonnegative solutions of the second. Thus the difference counts its solutions with at least one zero variable.

2. For $x = 0$, even $y = 0, 2, \ldots, 666$ gives 334; for $y = 0$, $x = 0, \ldots, 500$ gives 501; for $z = 0$, $y = 0, 4, \ldots, 664$ gives 167. Only the two axial solutions $(0, 0, 1000), (500, 0, 0)$ are double counted.

3. Difference $334 + 501 + 167 - 2 = 1000$, so remainder 0.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q06] | Part B (Q6-Q10)

Answer: 282

1. Before $500^2 = 250000$, successive squares differ by at most 999, so every earlier thousand-number block contains a square. It remains to check blocks starting with index 250.

2. For $x = 500 + a$, $x^2 = 1000(250 + a) + a^2$. For $0 \leq a \leq 31$, the final term is below one thousand, so blocks 250 through 281 all contain these squares.

3. The next squares are $531^2 = 281961$ and $532^2 = 283024$. They leave the entire block 282000 through 282999 empty of squares.

4. Therefore the least N is 282.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q07] | Part B (Q6-Q10)

Answer: 945

1. Let the starting staff be n and each removed group $k > 0$. Work balance gives $30n + 30(n-k) + 30(n-2k) + 5(n-3k) = 1775$, hence $19n - 21k = 355$.

2. Modulo nineteen, $k \equiv 3$, so $k = 3 + 19t, n = 22 + 21t$ with integer $t \geq 0$. The final staff is $n - 3k = 13 - 36t > 0$, forcing $t = 0$.

3. The first ninety-minute output is $30(22) + 15(19) = 945$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2015-II-Q08] | Part B (Q6-Q10)

Answer: 036

1. If either variable is 1, the target is 1. Otherwise both are at least 2, and the condition rearranges to $(2a-3)(2b-3)<5$.

2. The two factors are positive odd integers. Their only possibilities, apart from swapping, are $(1,1),(1,3)$, giving $(a,b)=(2,2),(2,3)$.

3. The target values are $65/16$ and $217/35=31/5$. The maximum is $31/5$, larger than 1, and $p+q=36$.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-II-Q06] | Part B (Q6-Q10)

Answer: 195

1. Write the square as m^2. Completing the square gives $(2m-(2n+85))(2m+(2n+85))=843=3\cdot 281$. The first factor is positive because $4m^2>(2n+85)^2$.

2. The increasing positive odd factor pairs are $(1,843)$ and $(3,281)$. If the factors are u,v, then $n=(v-u-170)/4$, giving168 and27, respectively; the corresponding m are211 and71.

3. Both satisfy the original equation and positivity. The requested sum is195.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q14] | Part C (Q11-Q15)

Answer: 108

1. Divide by seven. Only summands $1, 11, 111$ can occur, since $1111 > 1000$. If their multiplicities are x, y, z, then $x + 11y + 111z = 1000$, while $n = x + 2y + 3z = 1000 - 9(y + 12z)$.

2. For $z = 0, \ldots, 9$, the possible values $v = y + 12z$ lie in integer intervals with endpoints $(0, 90), (12, 92), (24, 94), (36, 96), (48, 98), (60, 100), (72, 102), (84, 104), (96, 106)$, and the single value 108.

3. The union is $\{0, 1, \ldots, 106, 108\}$, of size 108. Each gives a distinct positive n, so the answer is 108.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-II-Q13] | Part C (Q11-Q15)

Answer: 418

1. Put $S(x) = P(x) - x - 3$. Since $S(17) = S(24) = -10$, division by the two monic linear factors gives $S(x) = -10 + (x-17)(x-24)Q(x)$ with integer-coefficient Q.

2. An integer zero requires $(n-17)(n-24) \mid 10$. The factors differ by seven; testing signed factor pairs of divisors of ten leaves only $(2,-5)$ and $(5,-2)$, giving $n = 19, 22$.

3. The stipulated two distinct solutions must be these, so their product is 418.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-II-Q15] | Part C (Q11-Q15)

Answer: 181

1. For a positive candidate, write $n^2 = 3u^2 + 3u + 1$. Then $(2n-1)(2n+1) = 3(2u+1)^2$. These coprime odd factors force one square and the other three times a square. The lower factor cannot be three times a square, since the upper would be two modulo three. Hence $2n - 1 = k^2$ with odd $k > 0$.

2. Write $2n + 79 = d^2$. Subtracting yields $(d-k)(d+k) = 80$. The positive even factor pairs $2, 40$; $4, 20$; $8, 10$ give $k = 19, 8, 1$. Oddness leaves $k = 19, 1$, giving $n = 181, 1$.

3. The larger works: $181^2 = 105^3 - 104^3$ and $2 \cdot 181 + 79 = 21^2$. Thus the maximum is 181.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-I-Q14] | Part C (Q11-Q15)

Answer: 905

1. Let u_j count entries equal to j. Subtract the count equation from the first- and fourth-power equations to get $u_2 + 2u_3 + 3u_4 = 163$ and $15u_2 + 80u_3 + 255u_4 = 4395$.

2. Eliminating u_2 gives $5u_3 + 21u_4 = 195$. Hence u_4 is a multiple of five and $0 \le u_4 \le 9$, leaving $u_4 = 0, 5$. The corresponding count vectors are $(226, 85, 39, 0), (215, 112, 18, 5)$.

3. Their square sums are $917, 905$, so the minimum is 905.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2011-I-Q15] | Part C (Q11-Q15)

Answer: 098

1. Vieta gives $a+b+c=0$ and $ab+bc+ca=-2011$. After changing all signs if needed, choose $a \geq b \geq 0$, with $c=-(a+b)$. Then $a^2+ab+b^2=2011$.

2. Complete the square: $(2a+b)^2=8044-3b^2$. Since $3b^2 \leq 2011$, it suffices to test $0 \leq b \leq 25$. In this interval the right side is a square only for $b=10$, where it is $7744=88^2$.

3. Thus $a=39, c=-49$, giving $39+10+49=98$. These values satisfy both Vieta relations, so they are achieved by an integer m.

Source / 来源

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-I-Q13] | Part C (Q11-Q15)

Answer: 059

1. If $k^3 \le n < (k+1)^3$, the next candidate cube is $(k+1)^3$. The worst ratio in that block is $(1+1/k)^3$, decreasing with k.

2. For $m \ge 8$, every block works from $n = 1$, so $Q = 1$. For $4 \le m \le 7$, $n = 1$ fails, $2 \le n \le 7$ is covered by8, and blocks $k \ge 2$ have ratio at most $27/8 < 4$, so $Q = 2$.

3. For $m = 3$, $n = 8$ fails; $9 \le n \le 26$ is covered by27, and later ratios are at most $64/27 < 3$, so $Q = 9$. For $m = 2$, $n = 31$ fails; $32 \le n \le 63$ is covered by64, and later ratios are at most $125/64 < 2$, so $Q = 32$.

4. The sum is $32 + 9 + 4 \cdot 2 + 2010 = 2059$, with remainder59.

Source / 来源